

3_Plot

September 28, 2025

```
[4]: import streamlit as st
import pandas as pd
import plotly.express as px
from utils import load_data
```

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[5]: st.set_page_config(page_title="Plot", page_icon=" ", layout="wide") # Sets the
    ↪page configuration to use a wide layout for the chart.
st.title("Data plot (Plotly)") # Displays the main header
```

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[5]: DeltaGenerator()
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[6]: # Load data (cached in utils.load_data)
df = load_data() # Loads the full dataset,
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[7]: # Find a date/time-like column
date_cols = [c for c in df.columns if any(k in c.lower() for k in ["date",
    ↪"time", "datetime", "timestamp"])] # Finds column names that indicate time
    ↪series data.
date_col = date_cols[0] if date_cols else None
if date_col is None: # Checks if a primary time column was successfully
    ↪identified.
    st.error("No date/time column found - cannot build a time series.") #
    ↪Displays an error if the X-axis data is missing.
    st.stop() # Halts execution since the core plot cannot be generated.
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```
[8]: # Ensure datetime dtype
if not pd.api.types.is_datetime64_any_dtype(df[date_col]):
    df[date_col] = pd.to_datetime(df[date_col], errors="coerce") # Converts the
    ↪time column to the required datetime object type.
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[9]: # Numeric columns
num_cols = [c for c in df.columns if pd.api.types.is_numeric_dtype(df[c])] #
↳ Filters the columns to find all available numeric data for plotting.
if not num_cols:
    st.error("No numeric columns found to plot.") # Displays an error if there
↳ are no Y-axis.
    st.stop() # Stops execution.
```

```
[10]: # Select columns to plot
choice = st.selectbox("Select column(s) to plot", ["All columns"] + num_cols,
↳ index=0) # Creates an interactive dropdown menu for column selection.
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[11]: # Select month range
if "month" in df.columns and df["month"].notna().any(): # Checks for the
↳ pre-calculated 'month' column
    months = sorted(df["month"].dropna().unique().tolist())
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    default_month = months[0]
    start, end = st.select_slider("Select month(s)", options=months,
    ↪value=(default_month, default_month)) # Creates an interactive slider
    mask = (df["month"] >= start) & (df["month"] <= end)
    pdf = df.loc[mask].copy()
    subtitle = f"Months: {start} to {end}"
else:
    pdf = df.copy() # Uses the entire dataset if no 'month' column exists.
    subtitle = "All rows"

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[12]: # Build Plotly figure
if choice == "All columns": # Executes the logic for plotting multiple series.
    long_df = pdf[[date_col] + num_cols].melt(id_vars=date_col,
    ↪var_name="column", value_name="value") # Pivots the table from wide to long
    ↪format, which Plotly requires for multi-line plots.
    fig = px.line(long_df, x=date_col, y="value", color="column", # Creates the
    ↪Plotly line chart with multiple colored lines.
                  title=f"Time series - {subtitle}",
                  labels={date_col: "Date", "value": "Value", "column":
    ↪"Column"})
else: # Executes the logic for plotting a single series.
    fig = px.line(pdf, x=date_col, y=choice, # Creates a standard Plotly line
    ↪chart for the selected column.
                  title=f"Time series - {choice} - {subtitle}",

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labels={date_col: "Date", choice: "Value"})
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[13]: st.plotly_chart(fig, use_container_width=True) # Renders the Plotly figure
      ↪ within the Streamlit app, adjusting to the container width. allowing for
      ↪ fast reloading on subsequent accesses.
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